Laboratory 6: Bayes Decisions and Model Evaluation

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Contents

1 MVG classifier performance across applications **2**
1.1 Evaluation of the Applications 2
1.2 Equivalence of Costs and Effective Priors 2

2 Comparison of Gaussian Classifier Variants **4**
2.1 Analysis of the Three Applications 4
2.2 Extended Range Analysis (Bayes Error Plots) 5

Chapter 1

MVG classifier performance across applications

In this chapter, we analyze the performance of the baseline Multivariate Gaussian (MVG) classifier across five different applications. Each application is parameterized by a specific triplet (π_1, C_{fn}, C_{fp}), representing the prior probability of the authentic class, the cost of a false negative, and the cost of a false positive, respectively.

1.1 Evaluation of the Applications

Question: What do you obtain?

The evaluation of the Standard MVG classifier across the five specified applications yields the normalized Detection Cost Function (DCF_n) and the minimum DCF (min DCF) values shown in Table 1.1.

Application Triplet	π_1	C_{fn}	C_{fp}	**Actual DCF$_n$**	**Min DCF**
App 1 (Baseline)	0.5	1.0	1.0	0.140	0.130
App 2 (High Legit Prior)	0.9	1.0	1.0	0.400	0.342
App 3 (High Impostor Prior)	0.1	1.0	1.0	0.305	0.263
App 4 (Strong Security)	0.5	1.0	9.0	0.305	0.263
App 5 (Ease of Use)	0.5	9.0	1.0	0.400	0.342

Table 1.1: Performance of the Standard MVG classifier across five target applications.

1.2 Equivalence of Costs and Effective Priors

Question: Observe how the costs of mis-classifications are reflected in the prior: stronger security (higher false positive cost) corresponds to an equivalent lower prior probability of a legit user.

The cost of misclassifications mathematically mirrors a shift in the prior probabilities. This relationship is formalized by the effective prior ($\tilde{\pi}$), which translates any application triplet into an equivalent application assuming uniform unit costs ($C_{fn} = 1, C_{fp} = 1$):

$$\tilde{\pi} = \frac{\pi_1 C_{fn}}{\pi_1 C_{fn} + (1 - \pi_1) C_{fp}} \tag{1.1}$$

The empirical results in Table 1.1 perfectly demonstrate this theoretical property:

Stronger Security Scenario: Application 4 imposes a high cost on granting access to an impostor ($C_{fp} = 9.0$) with a uniform prior ($\pi_1 = 0.5$). The effective prior calculation yields $\tilde{\pi} = 0.10$. Application 3 utilizes a naturally low prior for authentic users ($\pi_1 = 0.1$) with uniform costs. Both applications evaluate to the exact same actual DCF_n (0.305) and minimum DCF (0.263). Demanding stronger security forces the decision threshold to become far more conservative, which is mathematically indistinguishable from assuming that genuine users are inherently rare.

Ease of Use Scenario: Application 5 imposes a high cost on rejecting a legitimate user ($C_{fn} = 9.0$), yielding an effective prior of $\tilde{\pi} = 0.90$. Application 2 explicitly sets the genuine user prior to $\pi_1 = 0.9$. Once again, both configurations produce identical actual DCF_n (0.400) and minimum DCF (0.342) metrics. Lowering the barrier for entry shifts the decision boundary identically to assuming the vast majority of all users are legitimate.

Chapter 2

Comparison of Gaussian Classifier Variants

This chapter expands our analysis to compare three distinct Gaussian classifier variants: the Standard MVG, the Tied Covariance MVG, and the Naive Bayes classifier. We evaluate these models across three specific applications represented purely in terms of their effective priors ($\tilde{\pi} = 0.1$, $\tilde{\pi} = 0.5$, and $\tilde{\pi} = 0.9$) with uniform costs.

Table 2.1 summarizes the Actual DCF_n and Minimum DCF extracted for these configurations.

Eff. Prior ($\tilde{\pi}$)	**Model**	**Actual DCF_n**	**Min DCF**	**Calib. Loss (%)**
0.5	Standard MVG	0.140	0.130	7.7%
	Naive Bayes	0.144	0.131	9.9%
	Tied MVG	0.186	0.181	2.7%
0.9	Standard MVG	0.400	0.342	17.0%
	Naive Bayes	0.389	0.351	10.8%
	Tied MVG	0.463	0.442	4.7%
0.1	Standard MVG	0.305	0.263	16.0%
	Naive Bayes	0.302	0.257	17.5%
	Tied MVG	0.406	0.363	11.8%

Table 2.1: Performance metrics and calibration loss (calculated as $\frac{\text{Actual}-\text{Min}}{\text{Min}}$) for the three models across different effective priors.

2.1 Analysis of the Three Applications

Question: Which models perform best?

The intrinsic discriminative power of a model is measured by its Minimum DCF. Based on the data in Table 2.1, the **Standard MVG** and the **Naive Bayes** classifiers perform the best. At $\tilde{\pi} = 0.5$, the Standard MVG achieves the lowest overall cost (0.130), closely followed by Naive Bayes (0.131). At $\tilde{\pi} = 0.9$, the Standard MVG again holds a slight edge (0.342 vs 0.351). At $\tilde{\pi} = 0.1$, Naive Bayes narrowly outperforms the Standard MVG (0.257 vs 0.263). The Tied MVG classifier is demonstrably the worst-performing model across all scenarios.

Question: Are the relative results of the models consistent over the different applications?

The relative performance rankings are largely consistent across the different applications. The Standard MVG and Naive Bayes classifiers consistently form the top tier of performance, always exhibiting very similar Minimum DCFs. While they swap the exact first and second place positions at the extreme prior of $\tilde{\pi} = 0.1$, the gap between them remains marginal. The Tied MVG classifier is absolutely consistent in its ranking, remaining the worst-performing model across all three effective prior configurations.

Question: Now consider also actual DCFs. Are the models well calibrated (i.e., with a calibration loss in the range of few percents of the minimum DCF value) for the given applications?

To evaluate calibration, we assess the calibration loss, which is the difference between the Actual DCF_n and the Minimum DCF. The models are relatively well-calibrated only at the uniform prior ($\tilde{\pi} = 0.5$). Here, the calibration losses are minimal: 7.7% for Standard MVG, 9.9% for Naive Bayes, and 2.7% for Tied MVG, falling within an acceptable range of a few percents of the minimum DCF value. However, the models are *not* well-calibrated at the extreme priors ($\tilde{\pi} = 0.9$ and $\tilde{\pi} = 0.1$). As seen in Table 2.1, the calibration losses jump to the 10%–17% range for the top-performing models, indicating a significant degradation in the reliability of the theoretical thresholds.

Question: Are there models that are better calibrated than others for the different applications?

Yes. Ironically, the worst-performing model (**Tied MVG**) is the best calibrated. Its calibration loss remains remarkably low at $\tilde{\pi} = 0.5$ (2.7%) and $\tilde{\pi} = 0.9$ (4.7%), and it exhibits the lowest calibration loss at $\tilde{\pi} = 0.1$ (11.8%). Between the two best-performing models, neither is strictly dominant in calibration; Naive Bayes is better calibrated at $\tilde{\pi} = 0.9$ (10.8% vs 17.0%), while Standard MVG is slightly better calibrated at $\tilde{\pi} = 0.1$ (16.0% vs 17.5%).

2.2 Extended Range Analysis (Bayes Error Plots)

To fully understand the behavior of the classifiers, we evaluated the Bayes Error plots over an extended range of prior log-odds from -4 to $+4$, capturing highly skewed and extreme applications. These plots are shown in Figures 2.1, 2.2, and 2.3.

Question: What do you observe?

Across all three classifiers (Figures 2.1, 2.2, and 2.3), the minimum achievable cost is lowest at the center (prior log-odds = 0, corresponding to a uniform effective prior of $\tilde{\pi} = 0.5$). As the applications become more extreme (moving towards -4 or $+4$), both the minimum DCF and the actual DCF increase. Most importantly, the gap between the theoretical optimal performance (Minimum DCF) and the empirical performance (Actual DCF) noticeably widens at these extremes.

Question: Are model rankings consistent across applications (minimum DCF)?

Yes, the model rankings are highly consistent. By observing the Minimum DCF (blue curves) across the entire range from -4 to $+4$, the Standard MVG and Naive Bayes classifiers consistently maintain a lower profile than the Tied MVG classifier. The Standard MVG and Naive Bayes models trace nearly identical paths across the entire spectrum, proving that their superiority over the Tied MVG is independent of the specific

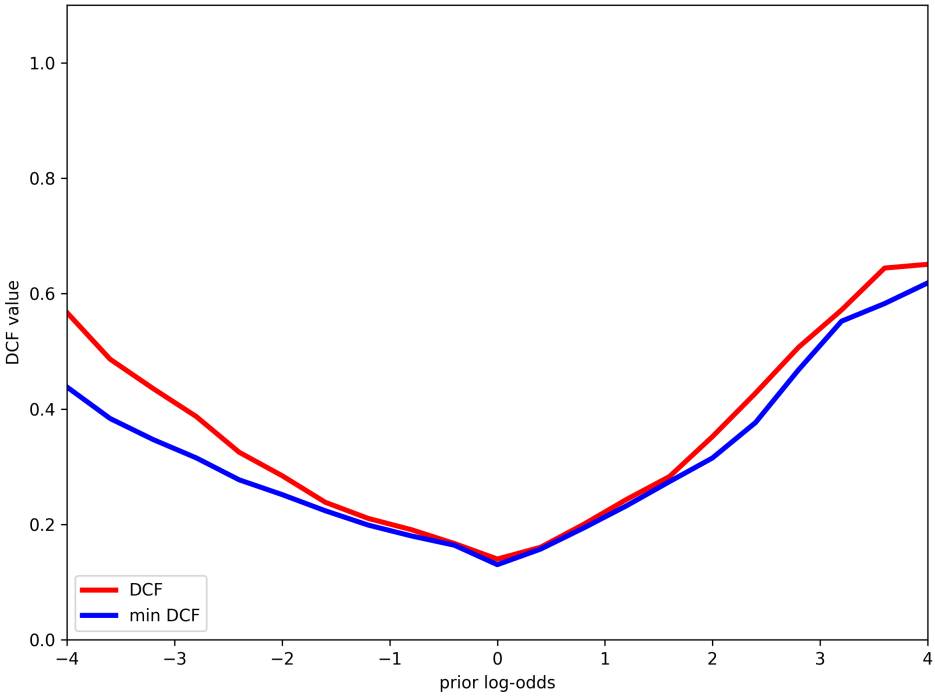


Figure 2.1: Bayes Error plot for the Standard MVG classifier. Note the increasing gap between the curves at extreme log-odds.

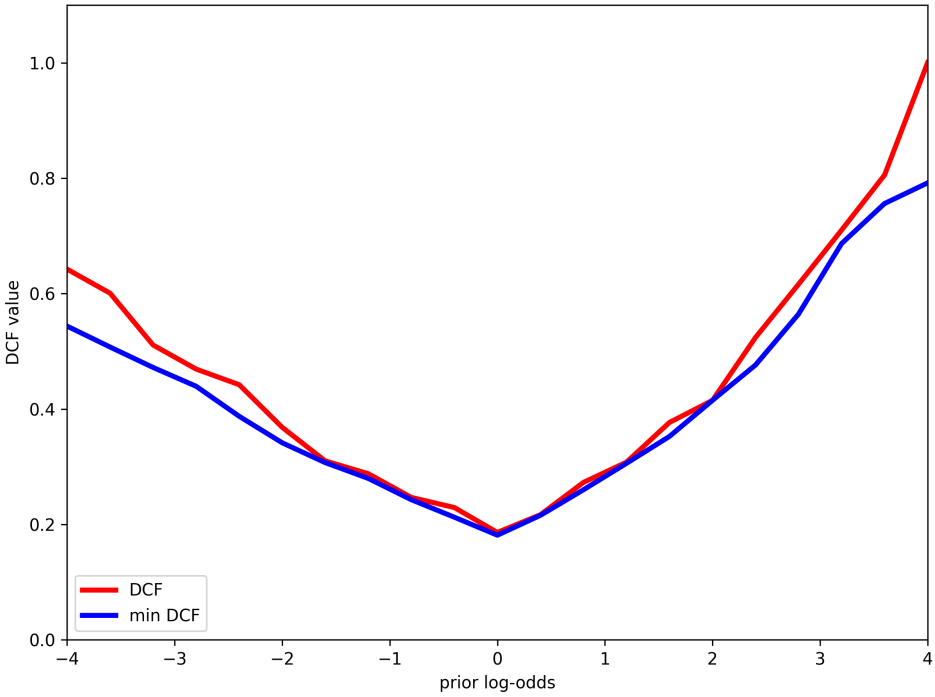


Figure 2.2: Bayes Error plot for the Tied MVG classifier. The curves remain closer together compared to the other models, indicating better relative calibration despite higher absolute costs.

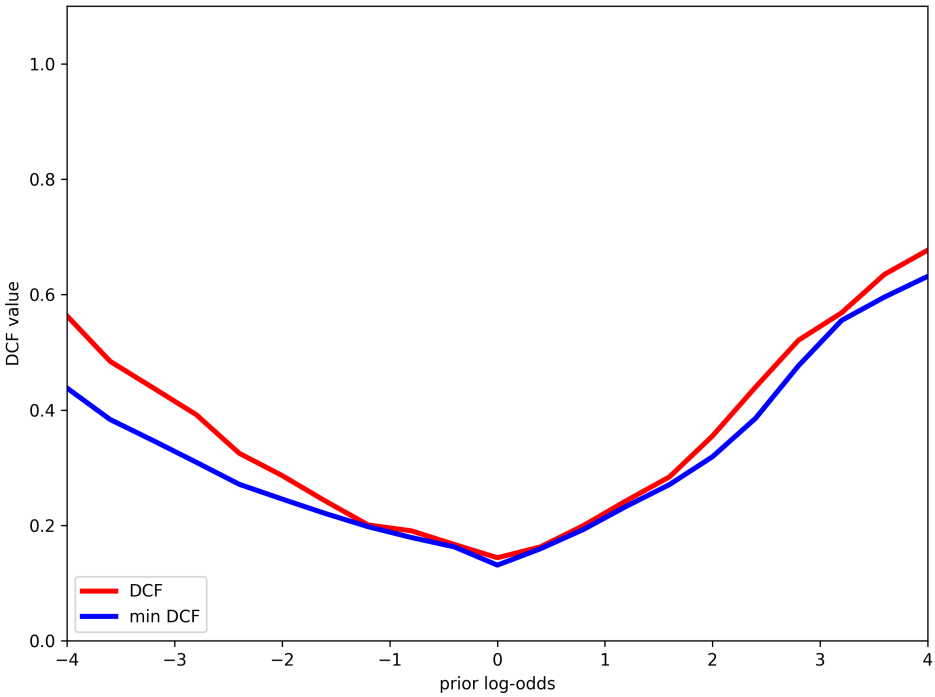


Figure 2.3: Bayes Error plot for the Naive Bayes classifier.

target application.

Question: Are models well-calibrated over the considered range of applications?

No, the models are not well-calibrated over the entire range. While they exhibit excellent calibration near log-odds = 0 (where the Actual DCF and Minimum DCF curves converge), they suffer from severe calibration loss at the extremes. This is most visibly prominent in the Tied MVG classifier (Figure 2.2), where at prior log-odds = +4, the Actual DCF reaches 1.0, while the Minimum DCF is approximately 0.8. The Standard MVG and Naive Bayes models also show a widening gap at the edges. This indicates that for highly unbalanced applications (e.g., severe class imbalance or highly asymmetric costs), the theoretical decision thresholds derived from the raw scores are sub-optimal, and explicit score re-calibration would be required to achieve the minimum theoretical cost.